

Nishanth Kasani

Dayton, OH | +1 (937) 516-7732 | nishanthknsn1@gmail.com | linkedin.com/in/nishanth-kasani |
nishanthknsn-mechanical.github.io/Nishanth.Portfolio

Professional Summary

Design Engineer with 2+ years of SolidWorks part, weldment, and assembly design, BOM development, ECO process execution, root cause analysis, prototype support, and cross-functional manufacturing collaboration at Cyient. Proficient in SolidWorks (CSWA Certified), CATIA V5, ANSYS, and MATLAB with background in hydraulic and fluid system analysis. M.S. Mechanical Engineering, University of Dayton. FE exam candidate.

Education

University of Dayton

Master of Science in Mechanical Engineering

Dayton, OH
Jan 2024 - May 2026 (Expected)

Chaitanya Bharathi Institute of Technology

Bachelor of Engineering in Mechanical Engineering

Hyderabad, India
Jun 2019 - May 2022

Professional Experience

Design Engineer

Aug 2021 – Nov 2023

Cyient

Hyderabad, India

- Utilized SolidWorks and CATIA V5 to design and detail parts, weldments, and assemblies, applying GD&T, datum callouts, and tolerance analysis, and maintained ECO/revision control and BOM accuracy across 50+ engineering orders.
- Developed and maintained Bills of Materials, executed ECO process for design changes, analyzed design and production issues using root cause analysis, and collaborated with manufacturing and vendor teams to implement practical solutions.
- Supported prototype builds and testing, documented test results and analysis, performed structural and thermal FEA using ANSYS and Abaqus, and coordinated with production personnel and technicians to resolve design issues.

Graduate Teaching Assistant

Aug 2024 - Present

University of Dayton

Dayton, OH

- Prepared composite and metal specimens for lab testing and operated equipment for cold working, heat treatment, and tensile, compression, and bending tests, ensuring accurate setup and safe machine operation.
- Collected and processed experimental data from mechanical tests, analyzed material property results, and maintained organized records to support coursework in materials science and mechanical systems.
- Guided 60+ undergraduate students through lab procedures, helping them understand material behavior, interpret test results, and follow safety protocols during hands-on sessions.

Projects

Reciprocating Saw: Weldment and Assembly Design with SolidWorks FEA | SolidWorks, ANSYS

- Utilized SolidWorks to design and detail a weldment and mechanical assembly, applied GD&T with weld callouts and machining tolerances for manufacturability, and supported prototype build and testing with complete BOM.
- Performed ANSYS FEA to validate structural performance under operating loads (Von Mises 87 MPa, FSF 2.3), documented test results and analysis, and executed design changes through ECO process.

Scroll Compressor: Hydraulic Fluid Pumping System Analysis | MATLAB, SolidWorks

- Analyzed a rotating fluid pumping system from first principles in MATLAB, modeling pressure-volume relationships, fluid flow, and volumetric efficiency directly analogous to hydraulic pump system design and performance analysis.
- Results matched published experimental data within 5%. Stander Symposium, April 2026. Demonstrates hydraulic and fluid system analytical depth applicable to fuel and hydraulic power system design. Manuscript in preparation.

Helical Coil Heat Exchanger: Hydraulic and Fluid System Design | ANSYS Fluent, SolidWorks

- Designed a fluid system and performed ANSYS Fluent CFD analysis confirming 20% improvement in fluid heat transfer and 12% pressure drop reduction, applying fluid dynamics principles relevant to hydraulic power system design.
- Developed complete SolidWorks model and technical documentation, validated performance within 3% of predictions, and demonstrated critical thinking in fluid system optimization.

Ladder Chassis Frame: Weldment Design, BOM, and Cost Improvement | SolidWorks, Abaqus

- Designed a structural weldment and assembly in SolidWorks, developed complete BOM, investigated and implemented a design change to improve cost effectiveness and manufacturability achieving 34.7% weight reduction.
- Validated through Abaqus FEA (FSF 2.8, deflection 3.1 mm), executed the design change through ECO process, and produced fabrication drawings with weld callouts and inspection criteria for production release.

Presentation: Kasani, N. (2026, April). *Variable Wall Thickness in Scroll Compressor*. Poster, University of Dayton Stander Symposium, Dayton, OH.

Technical Skills

CAD Software: SolidWorks (CSWA Certified, Simulation/FEA), CATIA V5, AutoCAD, Autodesk Inventor

Design: Weldment and Assembly Design, BOM Development, ECO Process, GD&T (ANSI/ASME Y14.5), DFM for Cost and Manufacturability

Fluid and Hydraulic Systems: Hydraulic Fluid System Analysis, MATLAB, ANSYS Fluent (CFD), Fluid Mechanics, Pump System Performance

Fabrication: Weld Design, CNC Machining, Sheet Metal, Composite Fabrication, FDM 3D Printing, Prototype Support

Other: Root Cause Analysis, Cross-Functional Collaboration, MS Office Suite (Excel, Word, PowerPoint), Python